ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

TECHNOLOGY

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 75

Marking guideline for the Academic Central Grade 7 Technology practice exam 3.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

QUESTION 1

1 1.1 A ✓ 1.2 B ✓ 1.3 D ✓ 1.4 A ✓ 1.5 C ✓ 1.6 C ✓ 1.7 A ✓ 1.8 D ✓ 1.9 B ✓ 1.10 D ✓ [10]

QUESTION 2

2.1 Natural frame, any ONE: a skeleton, a spider's web, a leaf's veins ✓. Manufactured shell, any ONE: a car body, a helmet, a cool-drink can, a bottle ✓ (2)

2.2.1 The tube ✓ (1)

2.2.2 Folding or rolling spreads the load over more material in depth (makes the beam deeper / stiffer) ✓, so the paper resists bending much better ✓. (2)

2.2.3 Any TWO: the same type and size of paper ✓; the same gap (20 cm) ✓; masses added in the same place and in the same way (2)

2.2.4 A bar graph ✓ (1)

2.3 The top of the beam is in compression ✓; the bottom of the beam is in tension ✓. (2)

2.4 Tubes are light ✓ but still strong and rigid, so the bicycle is easy to ride but does not bend ✓. (2)

2.5 Investigate ✓ → Design ✓ → Make ✓ → Evaluate ✓ → Communicate ✓ (5)

2.6 How the product looks (colour, shape, texture) and whether it is attractive ✓ (1)

[18]

QUESTION 3

3.1 (a) The ability to bend without breaking ✓ (b) The ability to resist scratches and dents ✓ (2)

3.2 Any suitable material, e.g. hardwood / chipboard with a laminated top / plastic laminate ✓, with a reason: strong, hard, smooth, scratch-resistant, easy to clean ✓ (2)

3.3 Reduce: use fewer materials, e.g. buy products with less packaging ✓. Reuse: use something again, e.g. use a glass jar as a container ✓. Recycle: process used materials into new products, e.g. melt glass bottles to make new bottles ✓. (3)

3.4 Trees are cut down and the logs are chipped ✓ → the chips are cooked with water and chemicals to make pulp ✓ → the pulp is spread out, pressed and dried into sheets of paper ✓ (3)

3.5 So that resources are not used up ✓ and the environment is protected for future generations ✓ (2)

[12]

QUESTION 4

4.1.1 Reverse-motion linkage ✓ (1)

4.1.2 To the left (the opposite direction) ✓ (1)

4.1.3 The output would move a larger distance ✓ (the input would move less than the output) (1)

4.2.1 $\text{MA} = 1{,}5 \div 0{,}5 \checkmark = 3 \checkmark$ (2)

4.2.2 $\text{Effort} = 1\,200 \div 3 \checkmark = 400\ \text{N} \checkmark$ (2)

4.2.3 $\text{MA} = 1{,}7 \div 0{,}3 \checkmark = 5{,}67 \checkmark$; less effort is needed ✓ (about 212 N) (3)

4.3 (a) Third class ✓ (b) Second class ✓ (2)

4.4 It tells us how many times a machine multiplies the effort ✓; the bigger the mechanical advantage, the less effort is needed to move the load ✓. (2)

4.5 Any TWO: oil or grease (lubricate) the moving parts ✓; use ball bearings ✓; make surfaces smoother; use wheels or rollers (2)

4.6 Any TWO: windscreen wipers ✓; bicycle brakes ✓; folding chairs or ironing boards; a toolbox with trays that open; a scissor lift; a car jack (2)

[18]

QUESTION 5

5.1 Correct symbols for the cell ✓, switch ✓, bulb and buzzer ✓; all components in one complete loop drawn with a ruler ✓ (4)

5.2.1 The more coils there are, the stronger the electromagnet ✓; it picks up about 3 or 4 more clips for every 10 extra coils ✓. (2)

5.2.2 About 17 or 18 paper clips ✓ (1)

5.2.3 The number of paper clips picked up ✓ (1)

5.2.4 Any ONE: disconnect the battery when not testing because the wire becomes hot; do not touch bare wire ends; do not connect the wire directly across the battery for long ✓ (1)

5.3 When the button is pressed, current flows through the coil and it becomes an electromagnet ✓. It attracts the iron arm, so the hammer strikes the gong ✓. This breaks the contact, the magnet switches off and the spring pulls the arm back; the circuit closes again and the process repeats quickly ✓. (3)

5.4 (a) A generator (dynamo) ✓ (b) A solar panel (solar cell) ✓ (2)

5.5 When current takes a path with almost no resistance (e.g. two bare wires touching) ✓; a very large current flows, which can overheat the wires, cause fires or damage the battery ✓. (2)

5.6 Light-emitting diode ✓ (1)

[17]

TOTAL: 75

END OF MARKING GUIDELINE